

CareHome Services

Technical Documentation

Full-Stack Web Application

React 18 * Spring Boot 3 * H2 / Oracle DB

Deployed on GitHub Pages + Cloudflare Tunnel

<https://adarshmalayath.github.io/Care-Project/>

1. Project Overview

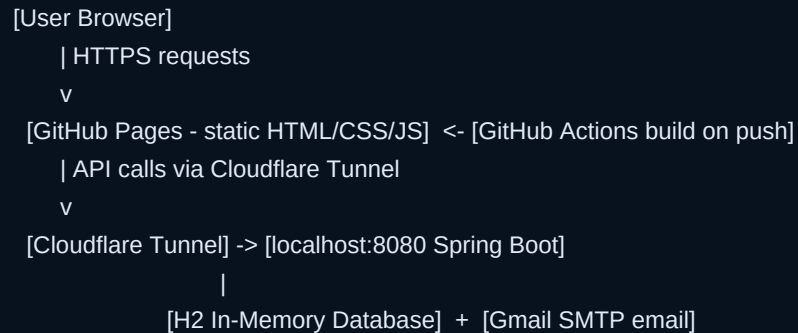
What is CareHome Services?

CareHome Services is a full-stack web application allowing families to submit care-home staffing enquiries (Care Workers, Drivers, Cooks, Other). Administrators manage all enquiries via a secure dashboard and reply via email.

Technology Stack

Frontend:	React 18, Vite, React Router v6, Framer Motion, Axios
Backend:	Spring Boot 3, Spring Security (JWT), Spring Mail
Database:	H2 In-Memory (dev) / Oracle (prod)
Styling:	Vanilla CSS with dark navy design tokens
CI/CD:	GitHub Actions -> GitHub Pages
API Tunnel:	Cloudflare Tunnel (exposes local backend)
Email:	Gmail SMTP via App Password (@Async)

System Architecture



2. User-Facing Pages

2.1 Homepage - Hero Section

Full-bleed hero with care-worker illustration, dark gradient overlay, and prominent CTA. The icon-only navbar (Home, Services, Admin Login) stays compact on all screen sizes.

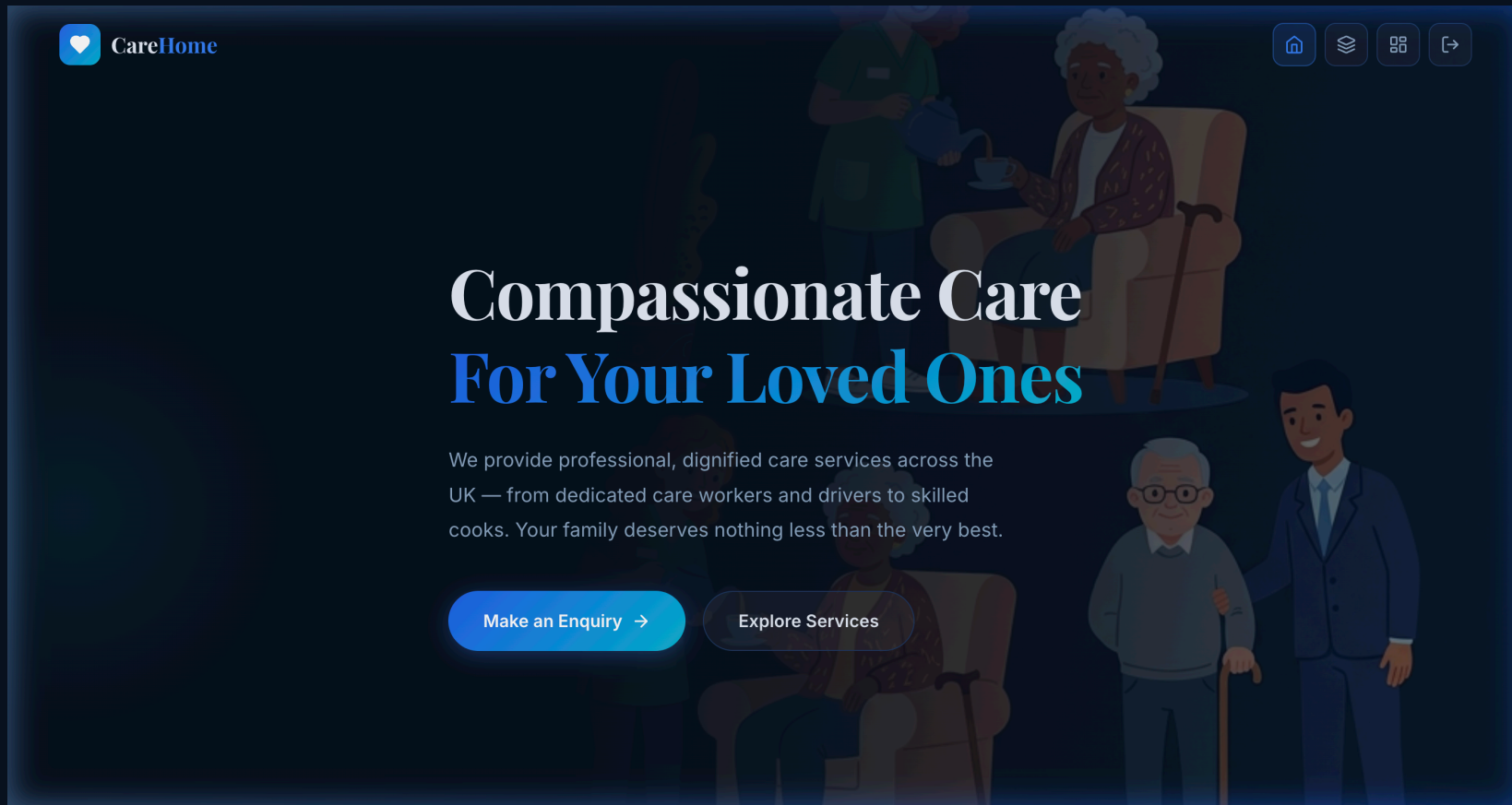


Fig 1: Homepage Hero Section

Four service cards in a responsive grid: Care Worker, Care Home Driver, Cook, Other Services. Each has an icon, description, and 'Enquire Now' button that opens the enquiry modal.

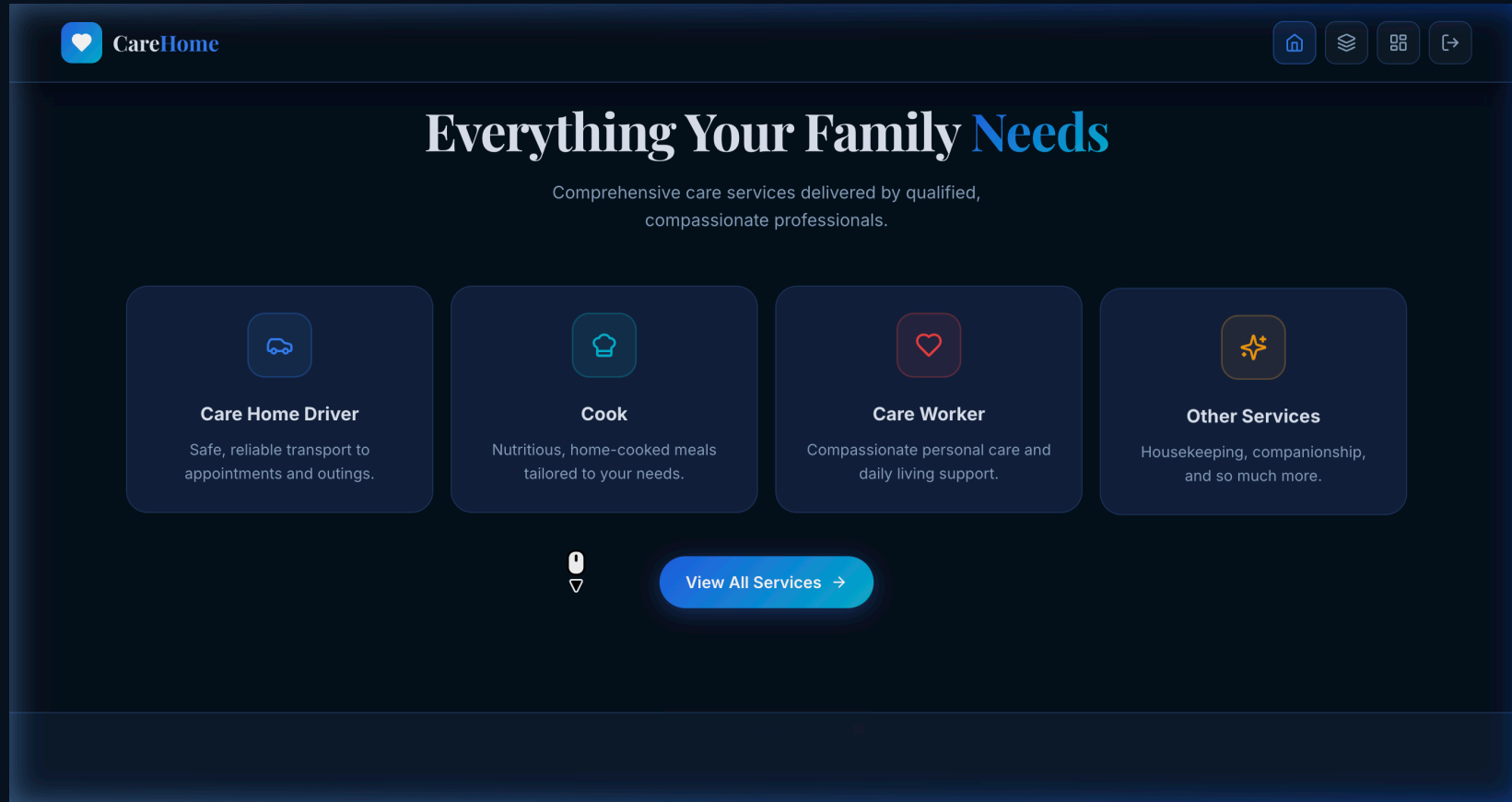


Fig 2: Services Cards on Homepage

Dedicated services page with expanded descriptions and a full enquiry form modal with validation, phone/address fields, and confirmation email on submission.

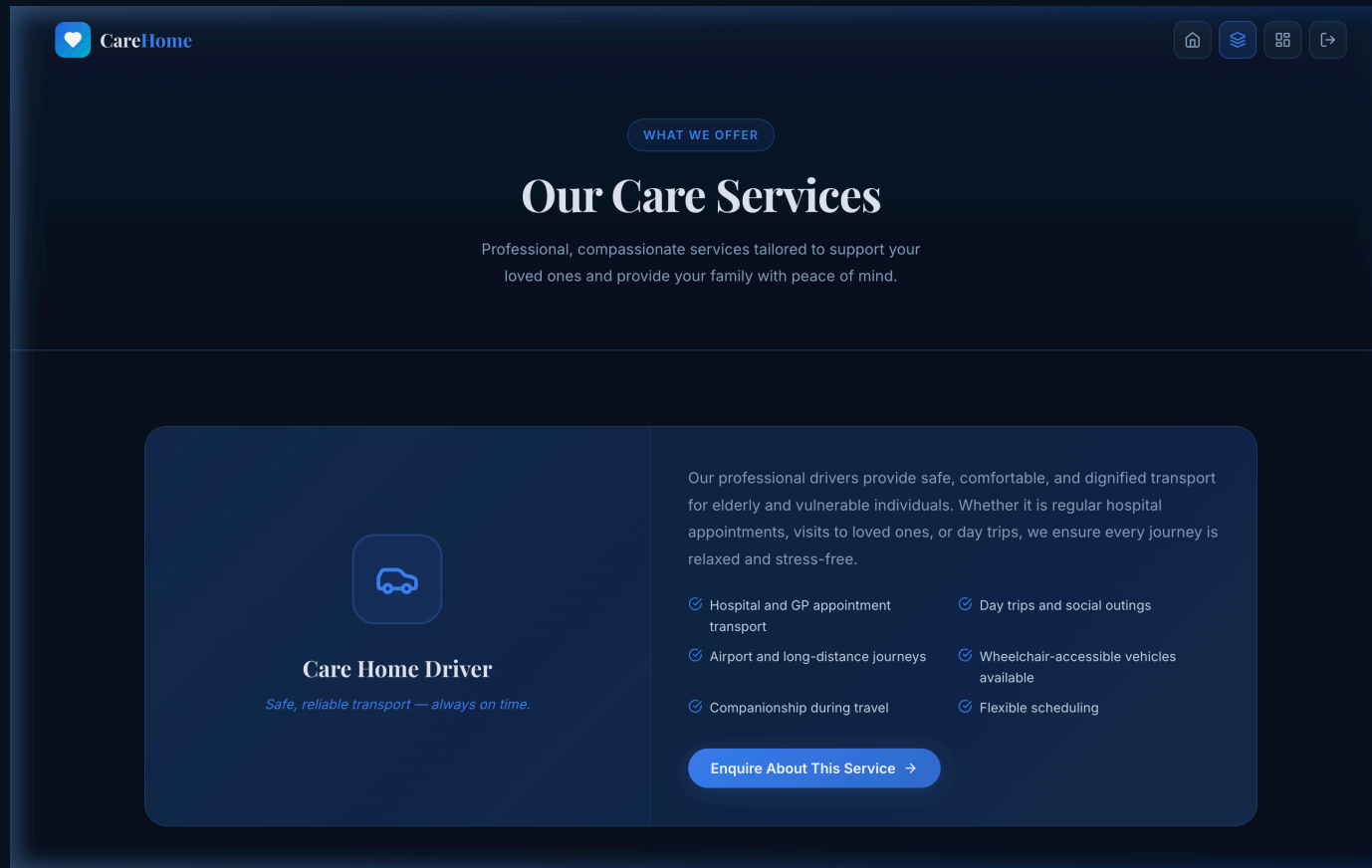


Fig 3: Full Services Page

3. Admin Module

3.1 Admin Login (/admin/login)

Secure login with username + password. On success, JWT is stored in AuthContext. Invalid credentials show error toast. Protected routes redirect unauthenticated users back here.

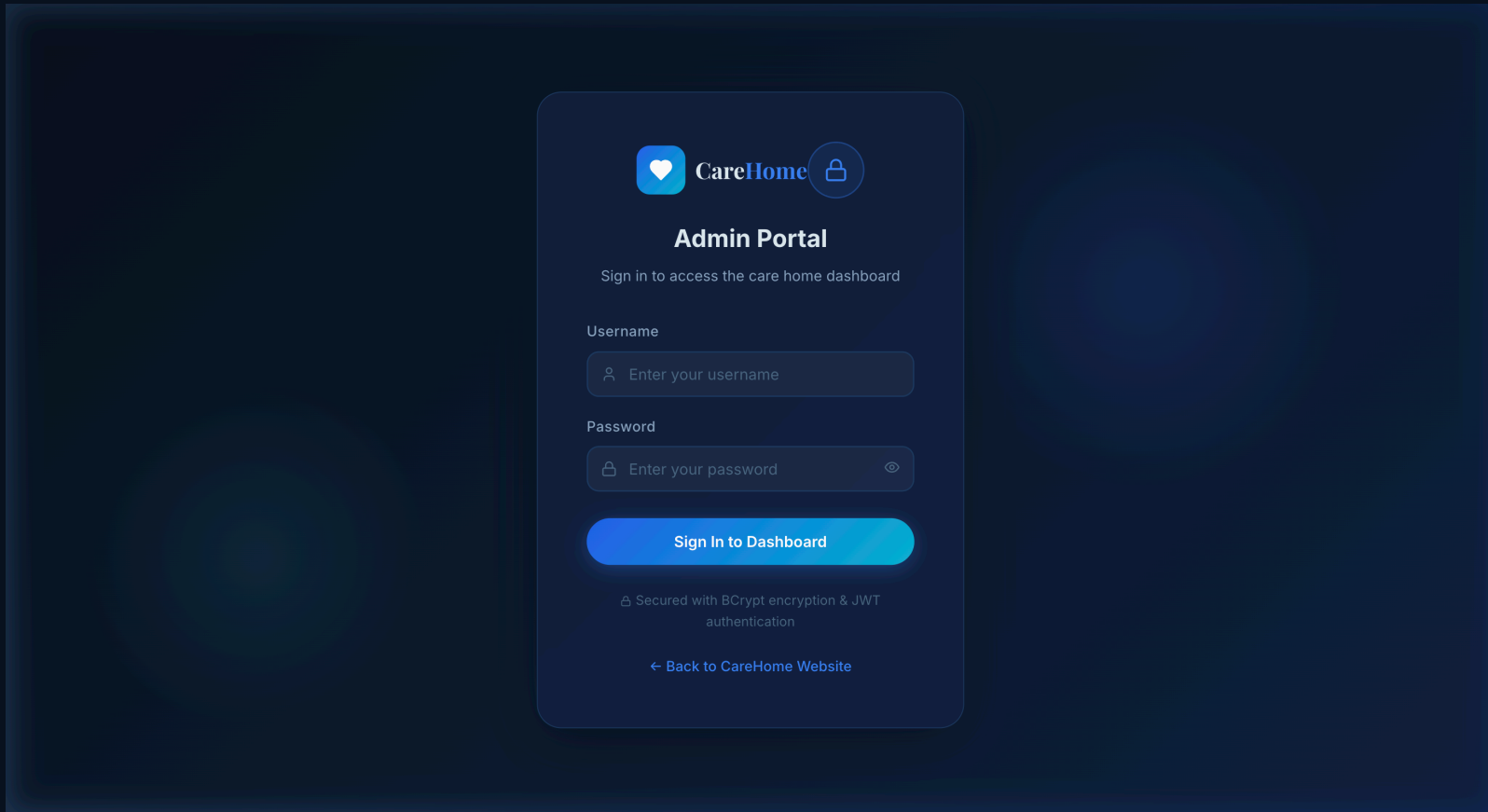
A screenshot of the Admin Login page. The page has a dark blue background. In the center, there is a white rounded rectangle containing the login form. At the top of this rectangle is the CareHome logo, which consists of a blue heart icon and the text 'CareHome' next to a blue padlock icon. Below the logo is the title 'Admin Portal' in bold. Underneath the title is the text 'Sign in to access the care home dashboard'. The form has two input fields: 'Username' with a placeholder 'Enter your username' and a user icon, and 'Password' with a placeholder 'Enter your password' and a password icon. Below these fields is a blue button with the text 'Sign In to Dashboard'. At the bottom of the form, there is a small text 'Secured with BCrypt encryption & JWT authentication' and a link '← Back to CareHome Website'.

Fig 4: Admin Login Page

Four stat cards (Total, Pending, Replied, Discarded) always show real totals regardless of active filter. Enquiry cards are filterable by status, searchable by name/email, and expandable for details.

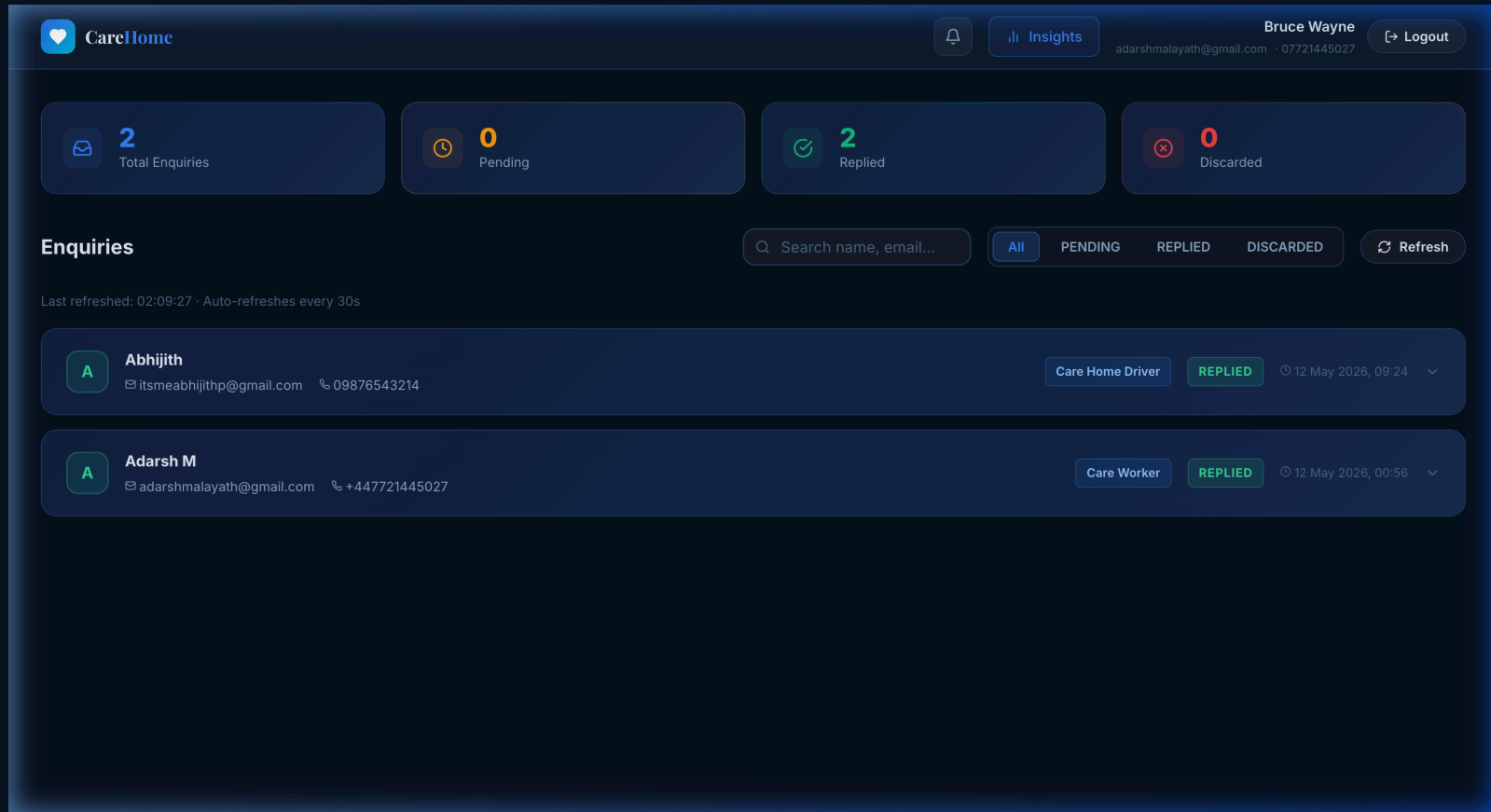


Fig 5: Admin Dashboard

The bell icon shows pending count badge. Clicking opens a slide-down overlay listing all PENDING enquiries with name, service, date, and a 'View' button that scrolls to that card without changing filter.

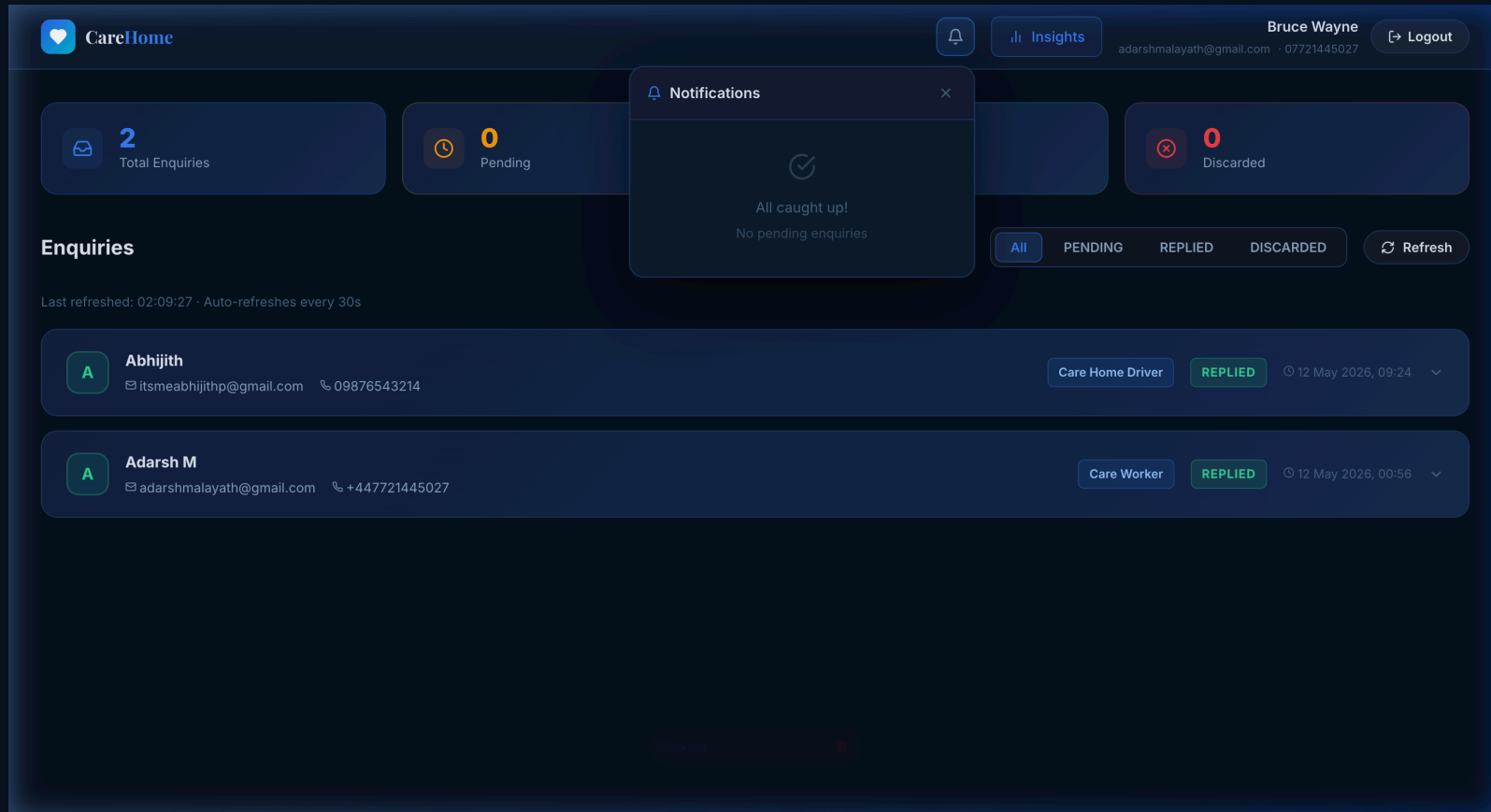


Fig 6: Notification Popup

Clicking a card expands to show address, message, admin reply (if any), and action buttons: Reply (sends email to customer) and Discard. Reply opens a modal with a text editor.

CareHome

Insights

Bruce Wayne

adarshmalayath@gmail.com · 07721445027

Logout

2

Total Enquiries

0

Pending

2

Replied

0

Discarded

Enquiries

Search name, email...

All

PENDING

REPLIED

DISCARDED

Refresh

Last refreshed: 02:09:27 · Auto-refreshes every 30s

A

Abhijith

itsmeabhijithp@gmail.com · 09876543214

Care Home Driver

REPLIED

12 May 2026, 09:24

ADDRESS

wembley

MESSAGE

"test 6"

ADMIN REPLY

anakk driverne tharaam

A

Adarsh M

adarshmalayath@gmail.com · +447721445027

Care Worker

REPLIED

12 May 2026, 00:56

Fig 7: Expanded Enquiry Card

4. Backend Architecture

4.1 Package Structure

/backend/src/main/java/com/carehome/

CareHomeApplication.java - Entry point with startup banner

config/

SecurityConfig.java - JWT + CORS + BCrypt (strength 12)

controller/

AdminController.java - /api/admin/** endpoints

EnquiryController.java - /api/enquiries endpoint

HealthController.java - /api/health ping

security/

JwtAuthFilter.java - Per-request token validation

JwtUtil.java - Token generation / parsing

service/

AdminService.java - Admin business logic

EnquiryService.java - Enquiry CRUD operations

EmailService.java - Async email via Gmail SMTP

4.2 Security Configuration

```
@Configuration @EnableWebSecurity
public class SecurityConfig {
    @Bean
    public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {
        http
            .csrf(AbstractHttpConfigurer::disable)
            .cors(cors -> cors.configurationSource(corsConfigurationSource()))
            .sessionManagement(s -> s.sessionCreationPolicy(STATELESS))
            .authorizeHttpRequests(auth -> auth
                .requestMatchers(POST, "/api/enquiries").permitAll()
                .requestMatchers(POST, "/api/admin/login").permitAll()
                .requestMatchers("/api/health").permitAll()
                .requestMatchers("/api/admin/**").hasRole("ADMIN")
                .anyRequest().authenticated()
            )
            .addFilterBefore(jwtAuthFilter, UsernamePasswordAuthenticationFilter.class);
        return http.build();
    }
    @Bean
    public PasswordEncoder passwordEncoder() {
        return new BCryptPasswordEncoder(12); // strength 12
    }
}
```


1. POST /api/admin/login -> AdminController validates credentials
2. BCrypt.matches(rawPw, hash) -> true
3. JwtUtil.generateToken(username) -> signed JWT (8hr expiry)
4. Response: { token, username, fullName, email }
5. Frontend stores token in AuthContext (memory, not localStorage)
6. Subsequent requests: Authorization: Bearer <token>
7. JwtAuthFilter extracts + validates token on every /api/admin/** call

4.4 Email Service (@Async)

```
@Service
public class EmailService {
    @Async
    public void sendEnquiryConfirmation(Enquiry e) {
        SimpleMailMessage msg = new SimpleMailMessage();
        msg.setTo(e.getEmail());
        msg.setSubject("Enquiry Received - CareHome Services");
        msg.setText("Dear " + e.getCustomerName() + ", your enquiry has been received...");
        mailSender.send(msg);
    }
    @Async
    public void sendAdminReply(Enquiry e, String replyText) {
        // sends reply email to customer
    }
}
```

5. API Reference

Base URL (production): <https://<cloudflare-tunnel>/api> Local: <http://localhost:8080/api>

Method	Endpoint	Auth	Description
POST	/enquiries	Public	Submit new customer enquiry (triggers confirmation email)
GET	/services	Public	List available service names
GET	/health	Public	Health check ping
POST	/admin/login	Public	Admin login - returns JWT token
GET	/admin/enquiries	JWT	List all enquiries, optional ?status= filter
POST	/admin/enquiries/{id}/reply	JWT	Reply to enquiry (emails customer)
POST	/admin/enquiries/{id}/discard	JWT	Mark enquiry as DISCARDED
GET	/admin/insights	JWT	Aggregated stats for insights charts

Request Body Example - POST /api/enquiries

```
{
  "customerName": "John Smith",
  "email": "john@example.com",
  "phone": "+44 7700 900000",
  "address": "12 Oak Street, London, SW1A 1AA",
  "serviceName": "Care Worker",
  "message": "Looking for a part-time care worker."
}
```

Response Example - POST /api/admin/login

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "username": "admin",
  "fullName": "Bruce Wayne",
  "email": "adarshmalayath@gmail.com"
}
```

6. Database Schema

6.1 ADMINS Table

```
CREATE TABLE ADMINS (  
  ADMIN_ID      BIGINT      NOT NULL DEFAULT NEXT VALUE FOR SEQ_ADMIN_ID,  
  USERNAME      VARCHAR(100) NOT NULL,  
  PASSWORD_HASH VARCHAR(255) NOT NULL,   -- BCrypt strength 12  
  EMAIL         VARCHAR(150),  
  FULL_NAME     VARCHAR(200),  
  PHONE         VARCHAR(25),  
  CREATED_AT    TIMESTAMP   DEFAULT CURRENT_TIMESTAMP,  
  IS_ACTIVE     BOOLEAN     DEFAULT TRUE,  
  CONSTRAINT PK_ADMINS PRIMARY KEY (ADMIN_ID),  
  CONSTRAINT UQ_ADMINS_USERNAME UNIQUE (USERNAME)  
)
```

6.2 ENQUIRIES Table

```
CREATE TABLE ENQUIRIES (  
  ENQUIRY_ID    BIGINT      NOT NULL DEFAULT NEXT VALUE FOR SEQ_ENQUIRY_ID,  
  CUSTOMER_NAME VARCHAR(150) NOT NULL,  
  EMAIL         VARCHAR(150) NOT NULL,  
  PHONE         VARCHAR(25)  NOT NULL,  
  ADDRESS       VARCHAR(500) NOT NULL,  
  SERVICE_NAME  VARCHAR(100) NOT NULL,  
  MESSAGE       CLOB,  
  STATUS        VARCHAR(20)  NOT NULL DEFAULT 'PENDING',  
  ADMIN_REPLY   CLOB,  
  REPLIED_BY    BIGINT,      -- FK -> ADMINS  
  CREATED_AT    TIMESTAMP   DEFAULT CURRENT_TIMESTAMP,  
  UPDATED_AT    TIMESTAMP   DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT PK_ENQUIRIES PRIMARY KEY (ENQUIRY_ID),  
  CONSTRAINT FK_ENQUIRIES_ADMIN FOREIGN KEY (REPLIED_BY) REFERENCES ADMINS(ADMIN_ID),  
  CONSTRAINT CHK_STATUS CHECK (STATUS IN ('PENDING','REPLIED','DISCARDED'))  
)
```

Indexes: IDX_ENQUIRIES_STATUS, IDX_ENQUIRIES_SERVICE, IDX_ENQUIRIES_CREATED

7. Frontend Architecture

7.1 Project Structure

/frontend/src/

App.jsx - Routes + BrowserRouter (basename=/Care-Project/)

index.css - Global design tokens + responsive media queries

api/

axios.js - Axios instance with VITE_API_URL base URL

services.js - Typed API call functions

context/

AuthContext.jsx - JWT auth state, login(), logout()

components/

Navbar.jsx - Icon-only nav with hover tooltips (Home,Services,Login)

Footer.jsx - Site footer

EnquiryModal.jsx - Customer enquiry form + validation

ReplyModal.jsx - Admin reply modal

ProtectedRoute.jsx - Redirects unauthenticated users to /admin/login

pages/

Home.jsx - Hero section + services overview

Services.jsx - Full services listing page

AdminLogin.jsx - Admin login form

AdminDashboard.jsx - Enquiry management + notification popup

AdminInsights.jsx - Analytics charts page

7.2 Key Design Decisions

- * Icon-only Navbar: saves mobile space, no hamburger menu needed
- * totalCount state: stat cards always show real totals, unaffected by active filter
- * SPA routing: 404.html redirect + index.html decode for GitHub Pages
- * Framer Motion: smooth card expand/collapse and notification popup animations
- * CSS custom properties: full dark navy design system with HSL tokens
- * @Async email: API never blocks waiting for email delivery

8. Deployment

8.1 GitHub Actions CI/CD

```
# .github/workflows/deploy.yml (simplified)
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: 20 }
      - run: npm ci          # install deps
        working-directory: frontend
      - run: npm run build    # vite build -> /dist
        working-directory: frontend
      env:
        VITE_API_URL: ${ secrets.VITE_API_URL }
      - uses: peaceiris/actions-gh-pages@v3
        with: { publish_dir: ./frontend/dist }
```

8.2 SPA Routing Fix for GitHub Pages

GitHub Pages returns 404 for any URL other than /. Fix: 404.html encodes the path as a query string and redirects to /. index.html decodes it back into browser history before React Router mounts.

8.3 Cloudflare Tunnel

The backend runs locally. Cloudflare Tunnel creates a public HTTPS URL:

```
cloudflared tunnel --url http://localhost:8080
```

This URL is set in frontend/.env.production -> VITE_API_URL, then a new build is deployed.

A new tunnel URL is generated each session (restart = update .env.production + redeploy).

8.4 Live URLs

Homepage:	https://adarshmalayath.github.io/Care-Project/
Services:	https://adarshmalayath.github.io/Care-Project/services
Admin Login:	https://adarshmalayath.github.io/Care-Project/admin/login
Admin Dashboard:	https://adarshmalayath.github.io/Care-Project/admin

8.5 Admin Credentials

```
Username : admin
Password : Admin@123
Email    : adarshmalayath@gmail.com
```